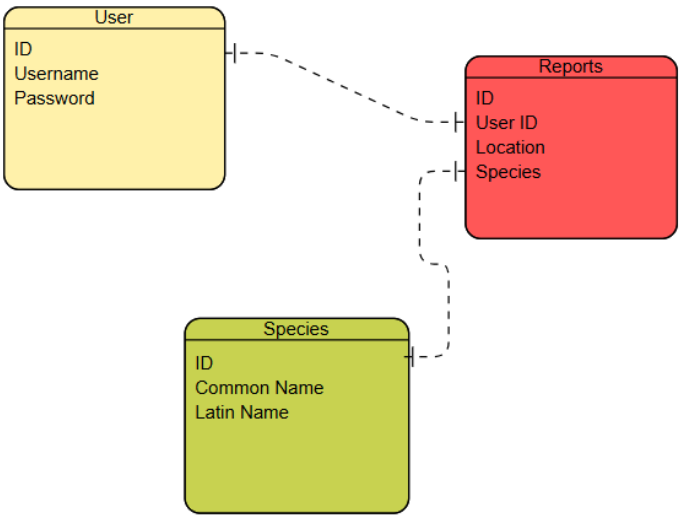
Web Application Development Log

Design Section

Project link on Github:
https://github.com/jackcado/yr12website-development-project

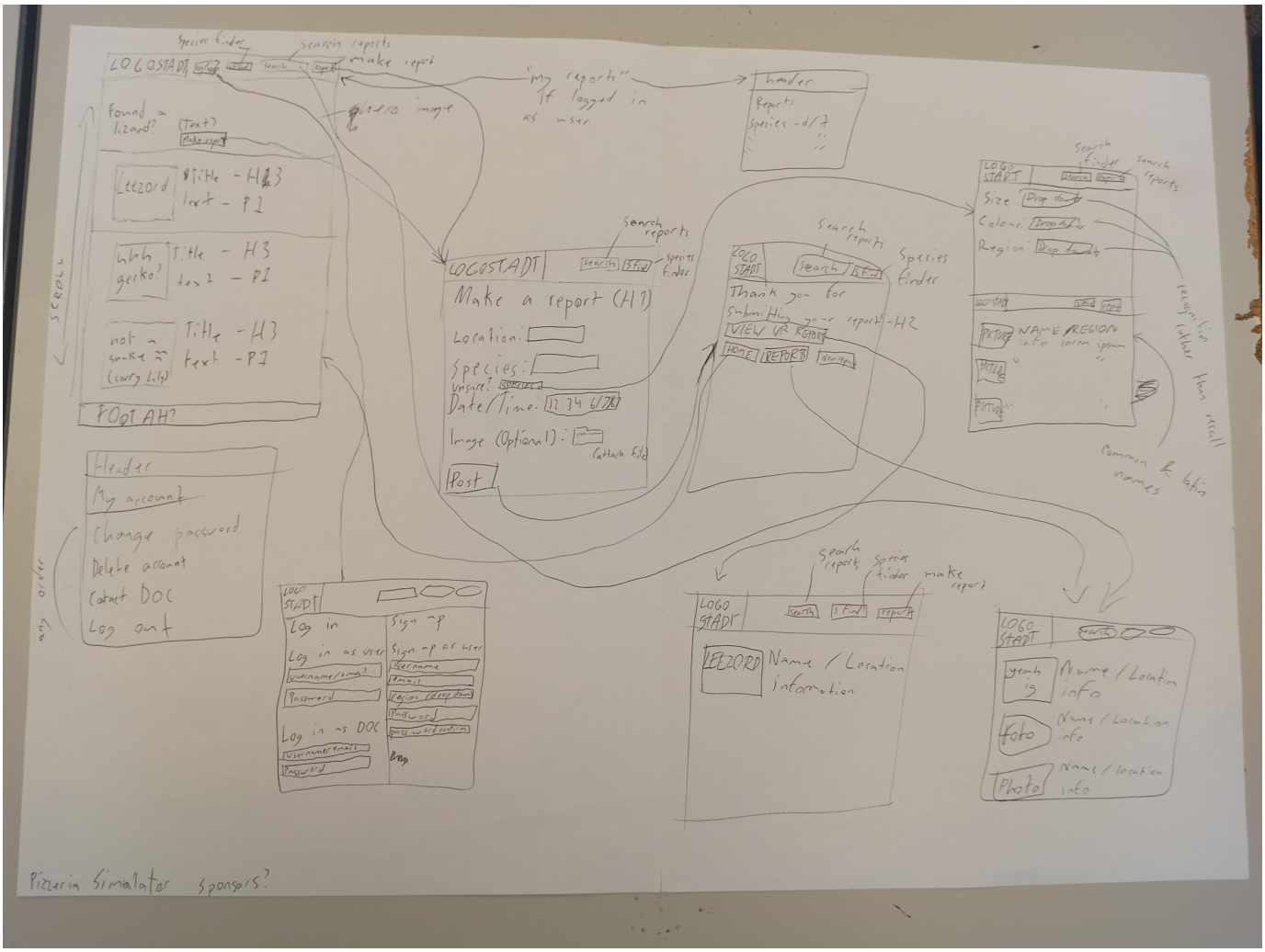
Project name:
Lizard Locator

Project Summary: Include purpose and target audience
The purpose of this website is to provide a simple and convenient way for people to report sightings of endangered lizard species to DOC directly. The target audience for this website are people who observe endangered species and wish to report them to DOC in a simple and easy format.

Initial Database Design ERD:


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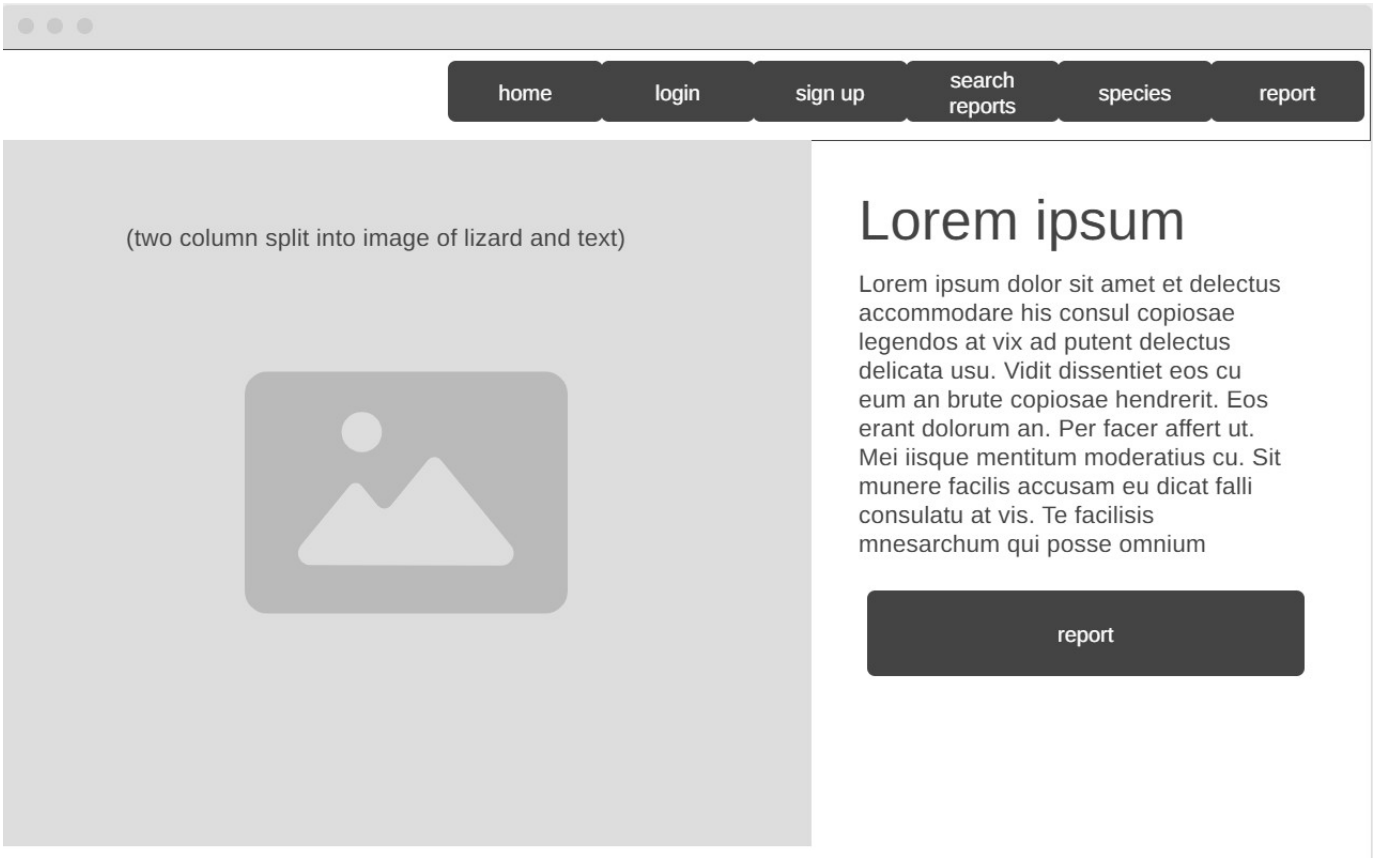
WebPage Design- Wireframe and/or Palette



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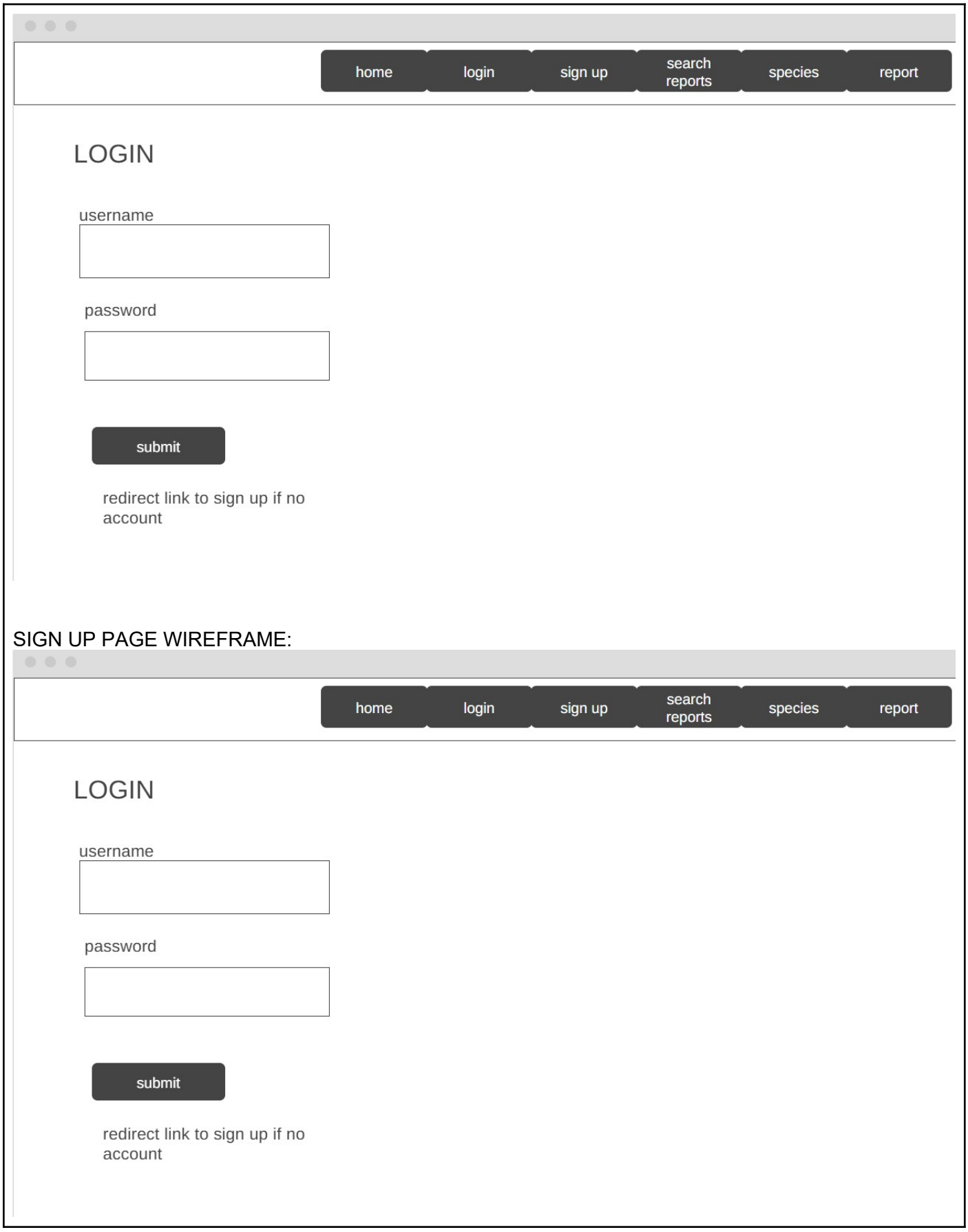
WebPage Design- Wireframe and/or Palette

INDEX PAGE WIREFRAME:



LOGIN PAGE WIREFRAME:

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SIGN UP PAGE WIREFRAME:

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Relevant Implications (explain at least 3)- Before Development.
Accessibility — I aim to make this website as accessible as I can to those who may struggle with certain website development choices, such as those who are visually impaired and use screen readers, colourblind individuals, and those with dyslexia. **Intellectual Property** — Only free to use assets will be used and credited in full in the README.md file. Please see file for full coverage on intellectual property compliance. **Privacy** — All passwords and personal user information will be hashed an unable to be accessed by even me.

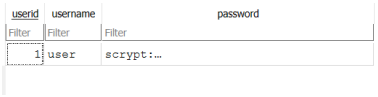
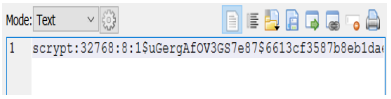
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Sprint #1

Sprint #1 Goals **(before starting sprint 1):**
- Complete ERD and wireframe - Have home page layout complete - Finish initial python script and templates - Mark up landing page - Working nav bar - Log in/Sign up working

Sprint #1 Testing

Include tests for everything in your application including all links, input forms, correctness of data and correctness of queries. Add more columns as required.

Testing Table			
What are you testing	**How are you testing it**	**Result**	**Pass/Fail**
All navigation bar buttons function	Click every link and see if it redirects to the right page	All redirect to the correct page	**PASS**
Sign up works	Insert username and password into sign up form and see if it adds it to the database	 New username/password in database	**PASS**
Password hashes correctly in database	Check database to ensure password is not in plain text	 Not in plain text	**PASS**
Log in with existing user	Insert username and password from the database into login form	Does absolutely nothing; could be flash error showing no result from form	**FAIL**
Log in with bad username/bad password	Make up a username/password that isn't in the database	Same result as logging in with existing user	**FAIL**

Sprint #1 Review

Progress Review (100+ words)

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Base built and functional for the most part. Finished base coding and styled partially. Navigation bar functional. Going to scrap ideas for footer at this point as not as important as other aspects and may not have enough time. Will come back to if have extra time after final testing for last sprint. Next sprint I will focus on fixing the flash that failed in this sprint and ensuring bad a username/password case feeds back to the user what the issue is via flash.

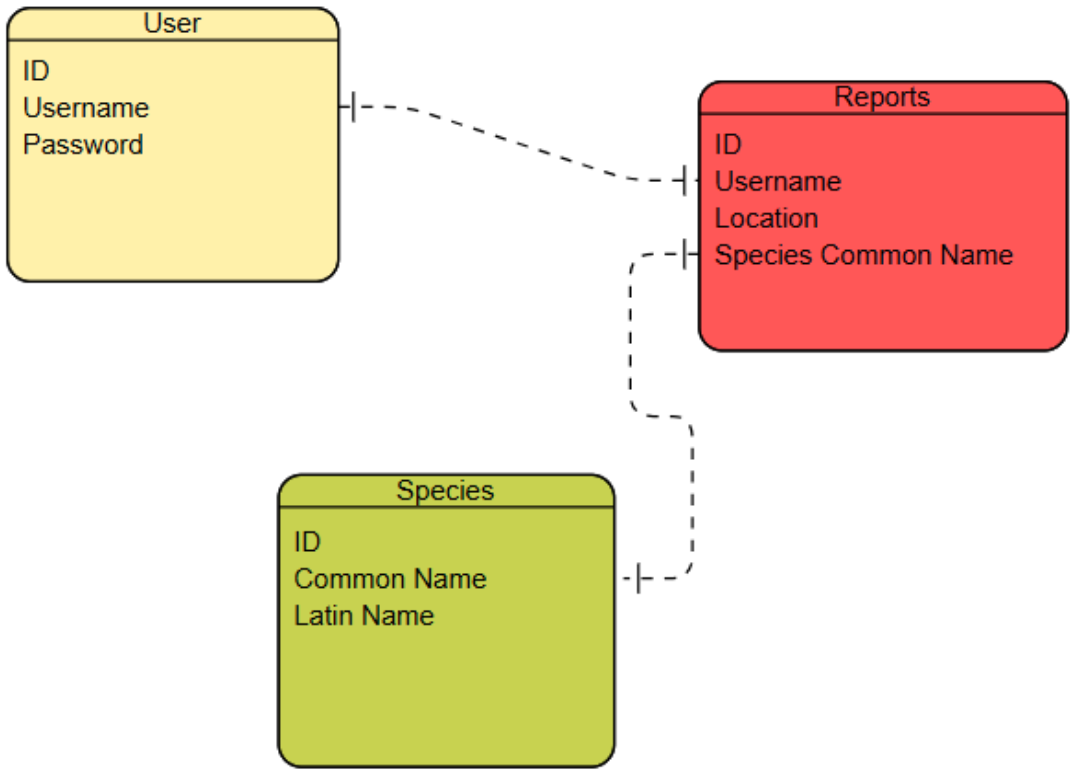
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Sprint #2

Sprint #2 Goals **(before starting sprint):**
- Fix bad username/password still working - Fix flash - Add post form for found reptile reports - Add list of species with photos to help users identify what they are reporting more accurately

Database improvements (if any)

Entity Relationship Diagram

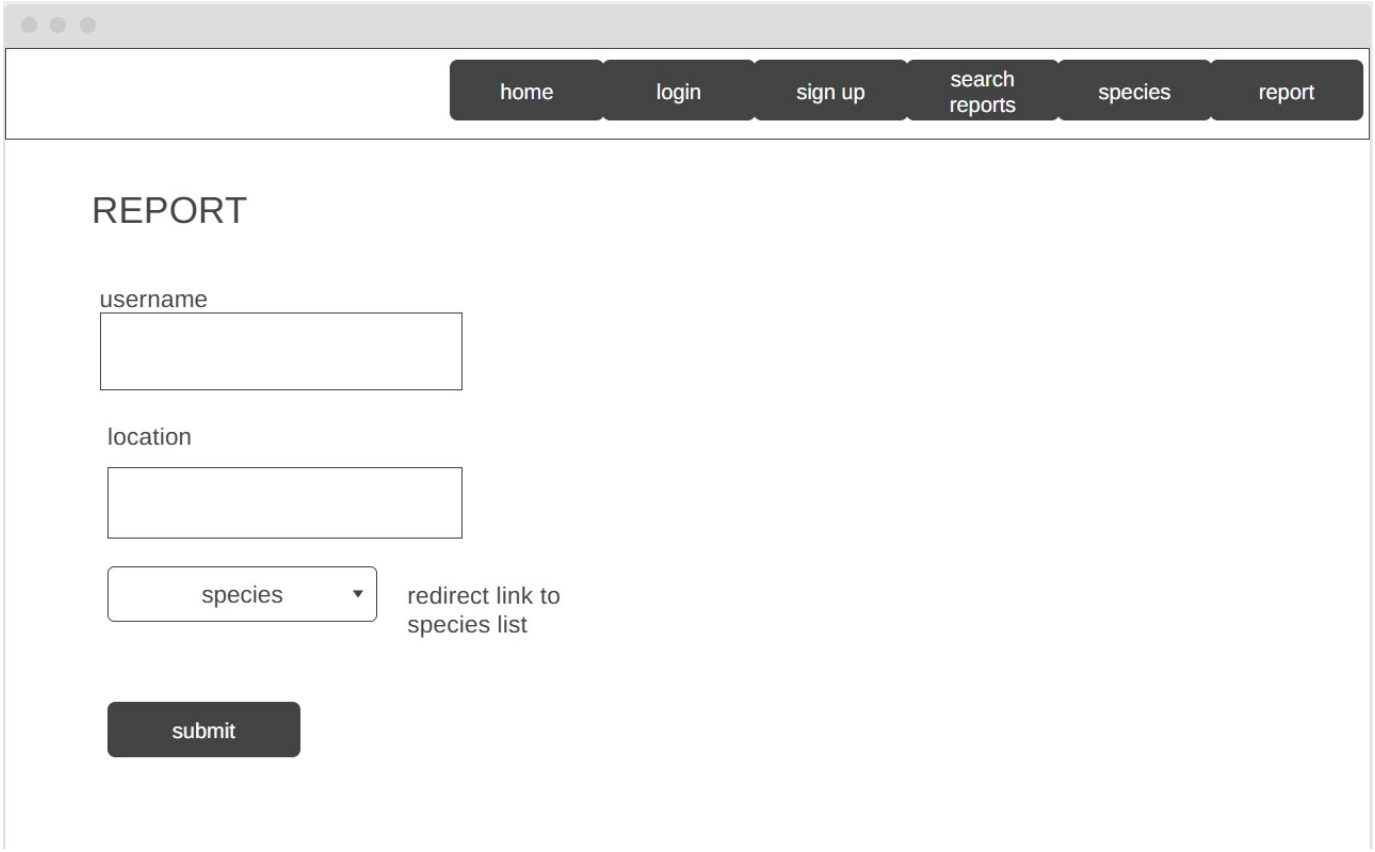


changed links to reports from other tables

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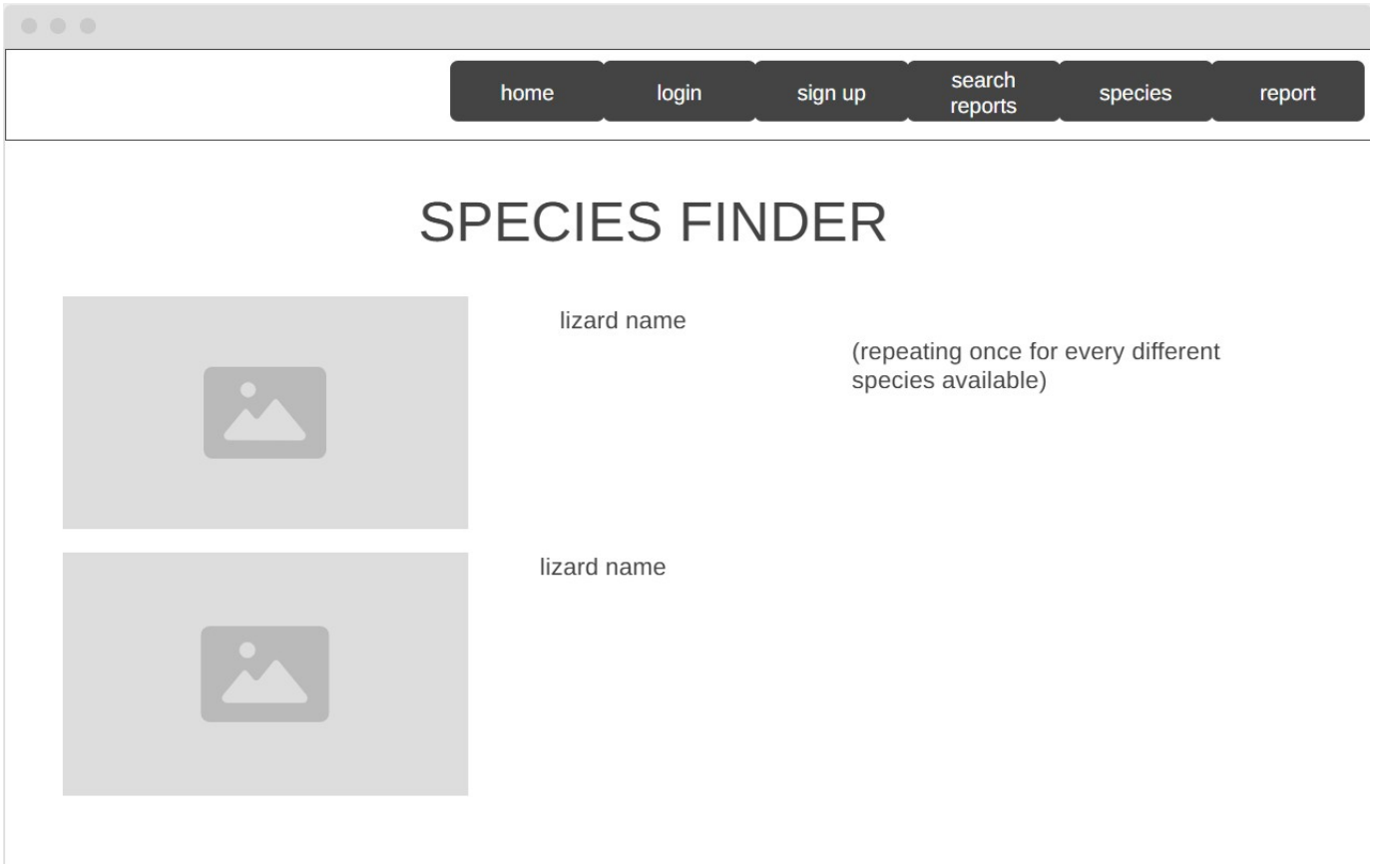
WebPage Design Improvements (if any)- Wireframe/Sketches/Pallette/Font etc

REPORT PAGE WIREFRAME:



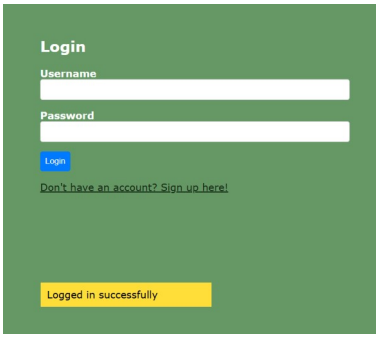
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SPECIES FINDER WIREFRAME:

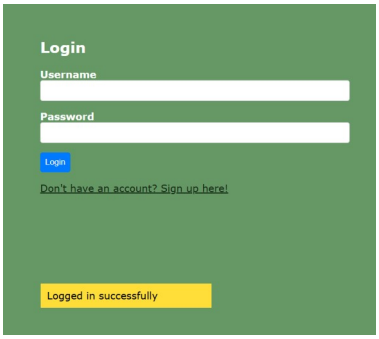
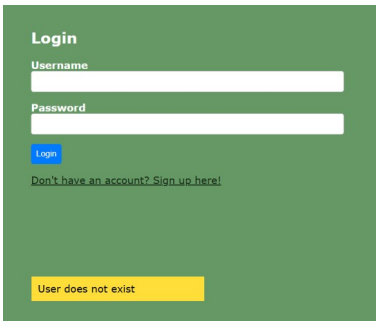
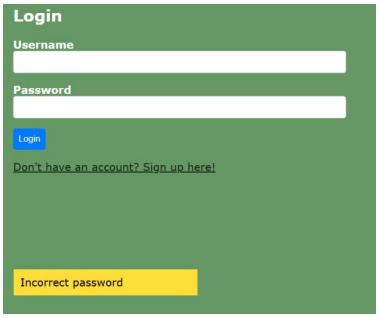
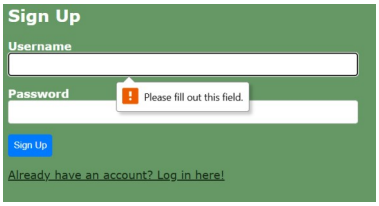
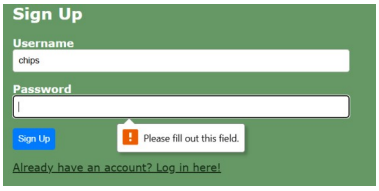


Sprint #2 Testing

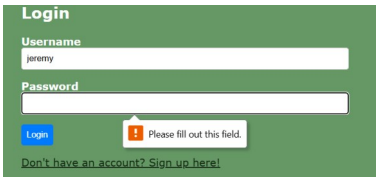
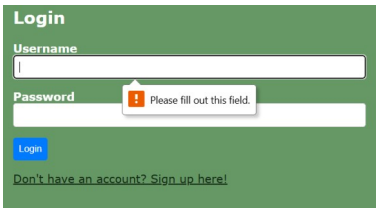
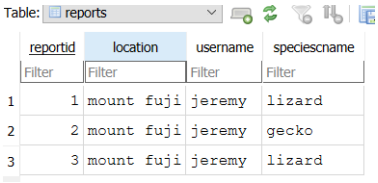
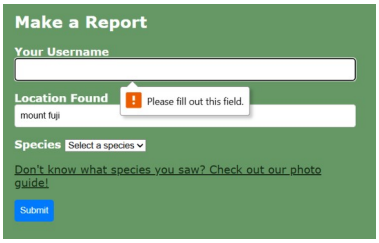
Include tests for everything in your application including all links, input forms, correctness of data and correctness of queries. Add more columns as required.

Testing Table			
What are you testing	**How are you testing it**	**Result**	**Pass/Fail**
Flash	Submit a login/sign up form and see if flash flashes	 Appears after submitting	PASS

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		login/sign up forms	
Log in with real user	Log in with correct username and password	 Logs in and flashes accordingly	PASS
Log in with non existent user	Log in with username that is not in the database	 Does not log in and flashes accordingly	PASS
Log in with bad password	Log in with username that is in the database, yet use incorrect password	 Does not log in and flashes accordingly	PASS
Blank forms	Fill in one or no boxes and hit submit	 	PASS

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		  User is instructed to fill in boxes and form is not submitted	
Report form functions	Fill in a report form and see if the information is entered into the database	 Data correctly inserted into database	PASS
Blank forms	Fill in one or no boxes and submit	   Form does not submit and flash tells the user what to do	PASS

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Sprint #2 Review

Progress Review (100+ words)
Flash is fixed and all aspects of login/sign up are functional. Added report form that adds to database. I do not like how the report form currently is. In next sprint I will change the form to take lizard species from database into drop down lists and remove username box and instead have it take from whatever username is stored in session. People can still put false locations into location but I can't really think of a way to stop that so all I can do is recognise it as a flaw and move on.

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Sprint #3

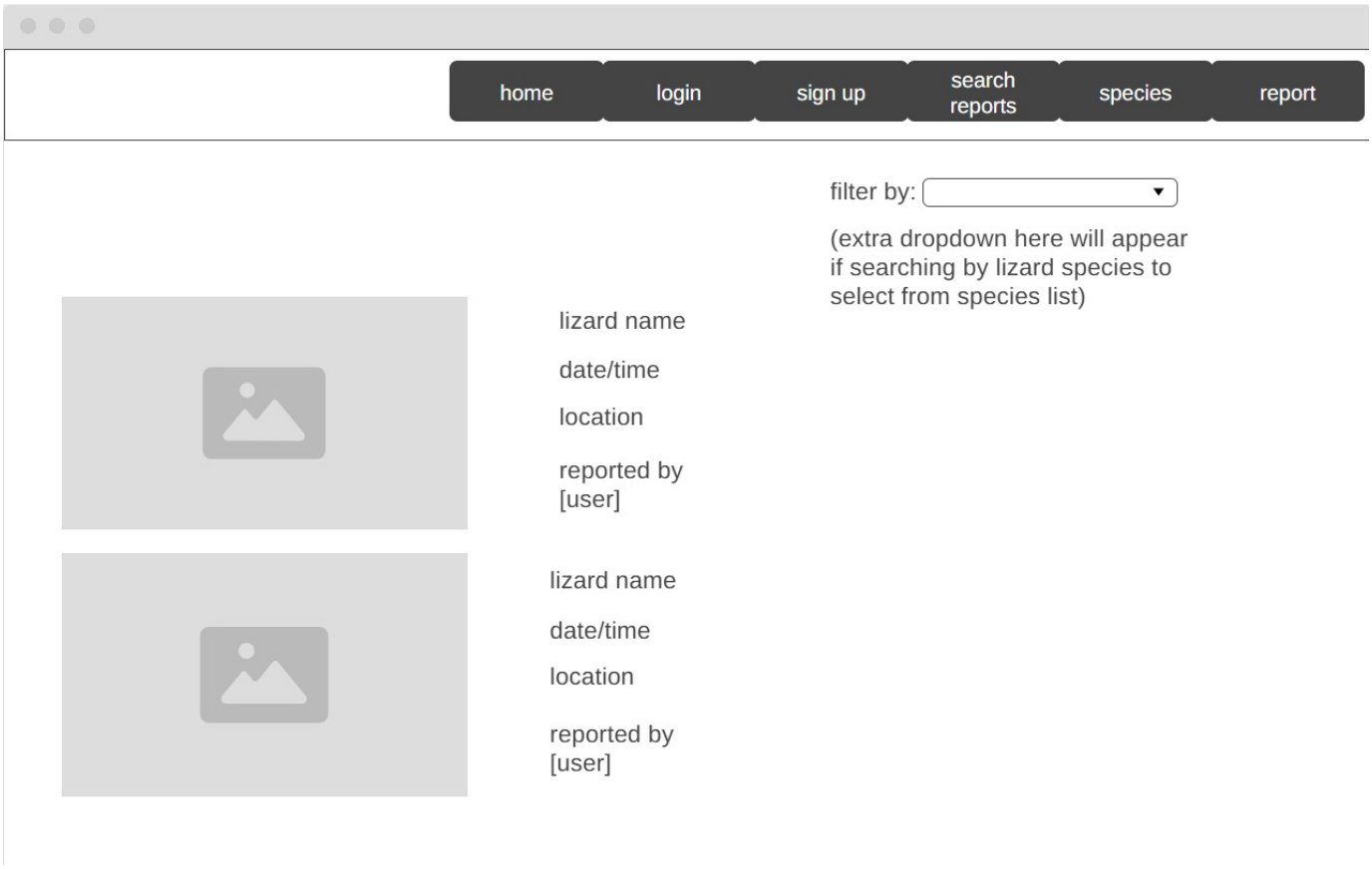
Sprint #3 Goals

(before starting sprint):

- Add search function so users can look through reports that have previously been made
- Change report form to take username from stored in session
- Change drop down menu in report form to take species names from the database

WebPage Design Improvements (if any)- Wireframe/Sketches/Pallette/Font etc

SEARCH REPORTS PAGE WIREFRAME:



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Sprint #3 Testing

Include tests for everything in your application including all links, input forms, correctness of data and correctness of queries. Add more columns as required.

Testing Table			
What are you testing	**How are you testing it**	**Result**	**Pass/Fail**
Unique usernames	Create a user with a name already existing in database.	Error. Did not create duplicate. Will fix and reattempt.	**FAIL**
Unique usernames (trial 2)	Create a user with a name already existing in database.	Did not create duplicate. User is asked to resubmit form with unique username using flash.	**PASS**
Reports	Fill out dummy report and ensure all info enters database.	All data found in database.	**PASS**
Search reports display	After filing report, ensure all info is displaying on page correctly.	All info displays correctly.	**PASS**
Image upload	Upload an image through report form and see if it appears in database correctly.	Appears in database, static upload folder.	**PASS**

Final Review

What were the results of your testing? Did you get through all that you expected to? What is causing problems? What went well? What was a challenge? What would you do differently if you had a chance?

Discuss how planning, testing and trialing led to the development of a high-quality digital technologies outcome.
I struggled a lot with login. Once I figured out that it was a database error I found the rest of the project made a lot more sense. I'm glad some things failed in my testing as it helped me identify and correct errors before final submission. If I had more time I would have liked to added a filter sort of function so users can search reports more efficiently, such as by species or location.

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How I addressed the Relevant Implications

As discussed before the project, what relevant implications had you identified at the beginning and how did you manage to address them as you went. Were there any others that you came across while making this game? How did you manage to address them? 100+ Words

Verdana (a proven dyslexic-friendly font) will be the font used throughout this website to assist those with dyslexia. Entire project has stuck to a consistent colour palate with readable font to ensure easy readability, colourblind friendliness and screen reader compatibility. Fonts are sized up from most standard websites to encourage readability as this was a concern one of my testers raised as she struggled reading the text due to it's size and her visual impairments. All assets used are free to use and come from Wikimedia Commons or are sourced from myself. Everything not mine is credited fully in my READ.md file. Privacy is upheld through having no important personal information (such as passwords) not being stored in plain text but by hashing and hash checking to protect in case of a breach. I would love to have added further security measures such as salt or pepper. Users are not asked to submit any private information that could be dangerous if breached.

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TECHQUITY

Marking Schedules: For Teacher use only

AS91892- Use advanced techniques to develop a database

Credits: 4 (internal)
NZQA: https://www.nzqa.govt.nz/nqfdocs/ncea-resource/achievements/2019/as91892.pdf

Achieved- Develop a digital outcome to manage data	**Evidence**	✔
designing the structure of the data	Made an ERD	
using appropriate tools and techniques to structure, organise, query and present data for a purpose and end user	Competently normalized data structure Multiple queries across linked tables Basic Python CRUD functionality with text input/output	
applying appropriate data integrity and testing procedures	Testing Logs. Do the queries work and return what is expected?	
Explaining relevant implications.	Relevant Implications Questions answered appropriately	
Merit- Develop an informed digital outcome to manage data		
using information from testing procedures to improve the quality of the database	Suggested improvements implemented- improve database design, add or refine entities or tables and relationships Made improved ERD(s)	
structuring, organising and querying the data logically	No problems with structure and datatypes in the table. No wildcard queries, JOIN's used efficiently. 2NF at least.	
addressing relevant implications.	The outcome addresses the relevant implications mentioned	
Excellence- Develop a refined digital outcome to manage data		
iterative improvement throughout the design, development and testing process	Improved functionality of database and refinements to structure of the data.	
presenting the data effectively for the purpose and to meet end-user requirements.	Very refined outcome providing all needed functionality	

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TECHQUITY

AS91893 - Use advanced techniques to develop a digital media outcome

Credits: 4 (Internal)
NZQA: https://www.nzqa.govt.nz/nqfdocs/ncea-resource/achievements/2019/as91893.pdf

Achieved	**Evidence**	✔
using appropriate tools and techniques for the purpose and end users	Coded a website in HTML and CSS. Looks and functions mostly like intended in the design.	
applying appropriate data integrity and testing procedures	It functions as intended and the right pages show with the correct data when the links are clicked	
using relevant conventions for the media type	Site has basic usability (HCI)	
explaining relevant implications.	Relevant conventions section completed adequately	
Merit		
using information from testing procedures to improve the quality of the outcome	Testing and sprint reviews show evidence of specific improvement made based on the testing. No lorem ipsum anymore!! Accurate Data	
applying relevant conventions to improve the quality of the outcome	Decent HCI. Eg. generally shows “consistency and standards”, and/or “help and documentation” by having an about page.	
addressing relevant implications.	Has met relevant implications like NOT used copyright material, addresses privacy by keeping data safe (eg hashed passwords) etc.	
Excellence		
iterative improvement throughout the design, development and testing process to produce a high-quality outcome	Must be high quality. No obvious errors or inaccurate data.	
using efficient tools and techniques in the outcome’s production.	Kept a good log, used github to keep backups and record of work, lots of commits on github.	

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TECHQUITY

AS91896 - Use advanced programming techniques to develop a computer program

Credits: 6 (Internal)
NZQA: https://www.nzqa.govt.nz/nqfdocs/ncea-resource/achievements/2019/as91896.pdf

Achieved	Evidence	✔
writing code for a program that performs a specified task	Applications works as intended.	
using advanced techniques in a suitable programming language	“creating methods, functions, or procedures that use parameters and/or return values” flask requires that “responding to events generated by a graphical user interface (GUI) ”- links in webpage run routes that do queries “using additional non-core libraries”- using Flask.	
setting out the program code clearly and documenting the program with comments	Some code comments, code layed out relatively cleanly with constants/variable at the top, routes and functions after.	
testing and debugging the program to ensure that it works on a sample of expected cases.	Testing table shows the basic web page functionality was tested.	
Merit		
documenting the program with appropriate names and comments that describe code function and behaviour	Good variable and function names. Good and plentiful code comments.	
following common conventions for the chosen programming language	PEP8 Followed. Use Pylint or similar and check linting.	
testing and debugging the program effectively to ensure that it works on a sample of both expected cases and relevant boundary cases.	Lots of testing including request for non-existent id returning an error 404 page.	
Excellence		
ensuring that the program is a well-structured, logical response to the specified task	Enhanced program, e.g. functions that aren’t routes to make the code cleaner (eg query_db). Neat and logical code.	
making the program flexible and robust	Great structure making code easy to extend. Unbreakable code. Eg. Error-404 and error 505 handlers	
comprehensively testing and debugging the program.	Unbreakable code with a lot of tests done for every testable element of the program including unexpected input like a non existent route or page returning custom error 404 handler	

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